

Appendix A: Python Code

Problem 1: Gimbal Degeneracy Sketch

Listing 1: Problem 1.py: generates the intermediate-rotation sketch in Fig. 1.

Problem 2: ZYZ Angles, Angle–Axis, and Quaternion

Listing 2: Problem 2.py: evaluates the ZYZ angles, angle-axis pair, and quaternion for Problem 2.

Problem 3: Quaternion to Rotation Matrix and Angle–Axis

Listing 3: Problem 3.py: converts each quaternion in Problem 3 to a rotation matrix and angle-axis pair.

Appendix B: Hand Calculations

Corrections made when typing up the hand calcs below:

1. **Problem 1(b), page 1:** the boxed yaw formula was written as $\phi = \text{atan2}\left(\frac{r_{21}}{c\theta_{\pm}}, \frac{r_{12}}{c\theta_{\pm}}\right)$, but r_{12} doesn't show up in the ϕ equation. The correct value comes from the first column, $r_{21} = s\phi c\theta$ and $r_{11} = c\phi c\theta$ (the same terms $c\theta = \sqrt{r_{11}^2 + r_{21}^2}$ above it is built from).
2. **Problem 3(d), page 6:** the (1,2) entry of the rotation matrix was written as -0.639 . Evaluating $2(q_1q_2 - q_0q_3) = 2((0.5928)(0.08311) - (0.5332)(0.5978))$ gives -0.539 , matching the neighbouring entries $(0.271, 0.797, 0.736, -0.418, -0.533, 0.620, 0.732, 0.283)$ and the boxed axis $\hat{\mathbf{k}} = [0.701 \ 0.098 \ 0.707]^T$ exactly.

Problem 1

$$Rot Z(\phi) Rot Y(\theta) Rot X(\psi)$$

$$\begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}$$

$$\begin{bmatrix} c\theta & s\theta s\psi & s\theta c\psi \\ 0 & c\psi & -s\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix}$$

a) $R(\psi, \theta, \phi):$

$$\begin{bmatrix} c\phi c\theta & c\phi s\theta s\psi - s\phi c\psi & c\phi s\theta c\psi + s\phi s\psi \\ s\phi c\theta & s\phi s\theta s\psi + c\phi c\psi & s\phi s\theta c\psi - c\phi s\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix}$$

b) $s\theta = -r_{31}$

$$c\theta = \sqrt{r_{11}^2 + r_{21}^2}$$

$$\sqrt{r_{11}^2 + r_{21}^2} = \sqrt{c\phi^2 c\theta^2 + s\phi^2 c\theta^2}$$

$$\therefore \theta = \arctan 2(-r_{31}, \pm \sqrt{r_{11}^2 + r_{21}^2})$$

$$\phi = \arctan 2\left(\frac{r_{21}}{c\theta_{\pm}}, \frac{r_{12}}{c\theta_{\pm}}\right)$$

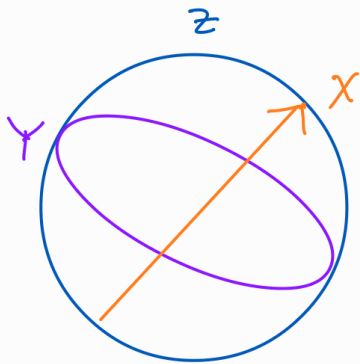
$$\psi = \arctan 2\left(\frac{r_{32}}{c\theta_{\pm}}, \frac{r_{33}}{c\theta_{\pm}}\right)$$

C) $\theta = \pi/2 + \pi n$ (90 degree Pitch)

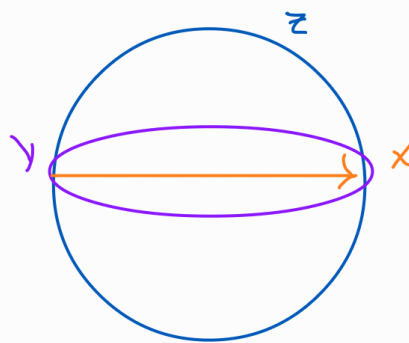
It becomes impossible to find their individual value, only the sum/diff.

D)

$\theta \neq 90$



$\theta = 90$



When you rotate pitch to 90° , it becomes parallel with the roll axis, so any rotation from then on spins both.

Problem 2

$$R = \begin{bmatrix} .6533 & .6403 & .4040 \\ -.3488 & .7282 & -.5900 \\ -.6719 & .2445 & .6991 \end{bmatrix}$$

a) $\det(R) = 1$ I used a calculator ☺

$$R \cdot R^T \approx \underline{\underline{I}}$$

b) for zyz :

$$\theta_{\pm} = \text{atan2}(\pm \sqrt{r_{13}^2 + r_{23}^2}, r_{33})$$

$$\phi = \text{atan2}\left(\frac{r_{23}}{\sin \theta_{\pm}}, \frac{r_{13}}{\sin \theta_{\pm}}\right)$$

$$\psi = \text{atan2}\left(\frac{r_{32}}{\sin \theta_{\pm}}, -\frac{r_{31}}{\sin \theta_{\pm}}\right)$$

$$\theta = 45.65^\circ \quad \phi = -55.6^\circ \quad \psi = 20.0^\circ$$

c) $\theta = \text{atan2}\left(\frac{1}{2} \left\| \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \right\|_2, \frac{\text{tr}(R) - 1}{2}\right)$

$$\hat{k} = \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

$$\therefore \theta = 57.3^\circ$$

$$\hat{k} = \begin{bmatrix} .496 \\ .639 \\ -.588 \end{bmatrix}$$

$$d) \quad q_0 = \pm \sqrt{\frac{1 + \text{tr}(R)}{4}}$$

$$q_1 = \frac{r_{32} - r_{23}}{4q_0}$$

$$q_2 = \frac{r_{13} - r_{31}}{4q_0}$$

$$q_3 = \frac{r_{21} - r_{12}}{4q_0}$$

$$Q = \begin{bmatrix} 878, [238, .307, -.282]^T \end{bmatrix}$$

Problem 3

a) $R([1, [0, 0, 0]^T]) =$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\theta = 0^\circ$$

$$\hat{k} = \vec{0}$$

b) $R([0, [0, 0, 1]^T]) =$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\theta = 180^\circ$$

$$\hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C) R([0, [1, 0, 0]^T]) =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \theta = 180^\circ$$

$$\hat{k} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$D) R([.5332, [.5928, .08311, .5978]^T]) =$$

$$\begin{bmatrix} .271 & -.539 & .797 \\ .736 & -.418 & -.533 \\ .620 & .732 & .283 \end{bmatrix} \quad \theta = 115^\circ$$

$$\hat{k} = \begin{bmatrix} .701 \\ .098 \\ .707 \end{bmatrix}$$

problem 4

$$R_K(\theta) = \underline{I} \cos \theta + [K]_x \sin \theta + K K^T (1 - \cos \theta)$$

$$\frac{dR}{dt} = \left(\frac{d}{d\theta} {}^0 R_i(\theta, \hat{K})^i R_{obj} \right) \dot{\theta} + \left(\frac{d}{d\hat{K}} {}^0 R_i(\theta, \hat{K})^i R_{obj} \right) \dot{\hat{K}}$$

$$\hat{K} = \frac{{}^0 w}{\|w\|_2}, \quad \frac{dw}{dt} = 0 \therefore \dot{\hat{K}} = 0$$

$$\theta = \|w\|_2 t \quad \therefore \frac{d\theta}{dt} = \|w\|_2$$

From $R_K(\theta)$

$$\frac{d}{d\theta} (R_K(\theta)) =$$

$$-\underline{I} S\theta + [K]_x C\theta + K K^T S\theta$$

$$\left. \frac{d}{d\theta} (R_K(\theta)) \right|_{\theta=0} = [K]_x$$

$$\frac{dR}{dt} = \left(\frac{d}{d\theta} {}^0R_i(\theta, K) {}^iR_{obj} \right) \| {}^0w \|_2$$

$$[K]_x = [K_x] \cdot {}^0R_{obj} \cdot \| {}^0w \|_2$$

$$\| {}^0w \|_2 \cdot [K_x] \cdot {}^0R_{obj}(0) = {}^0R_i(0, \hat{K}) {}^iR_{obj}$$

$$\| {}^0w \|_2 \cdot \left[\frac{w}{\| {}^0w \|_2} \right]_x \cdot {}^0R_{obj}(0) = {}^iR_{obj}$$

$$= [w]_x$$

$$\frac{dR}{dt} = [w]_x \cdot {}^0R_{obj}$$

Problem 5

$$a) \quad {}^0T_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3 =$$

$$\begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 2.5 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$b) \quad {}^3d_{32} = [-.5, .5, 3]^T$$

$$\begin{bmatrix} 0 \\ \vec{d}_{02} \\ 1 \end{bmatrix} = {}^0T_3 \cdot \begin{bmatrix} \vec{{}^3d_{32}} \\ 1 \end{bmatrix}$$

$${}^0d_{02} = \begin{bmatrix} -.5 \\ 2 \\ 1 \end{bmatrix}$$